Republic of the Philippines

Western Mindanao State University

College of Computing Studies

Department of Information Technology

In Partial Fulfillment of the Requirements for the Degree Bachelor of Science in Information Technology

Website:

EDU SOLVE

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CHAPTER 1

INTRODUCTION

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he Philippine education system stands at a critical juncture, grappling with persistent challenges in improving students' proficiency in essential academic subjects such as Mathematics, Science, and Reading. These challenges were starkly highlighted in the 2022 PISA (Programme for International Student Assessment) results, where the Philippines ranked among the lowest-performing countries globally. Such outcomes underscore systemic deficiencies and the urgent need for innovative solutions that transcend traditional teaching methods to bridge deep-rooted learning gaps.

EDU SOLVE rises as a transformative response to this pressing educational crisis. Tailored specifically for Grade 10 students, this web-based platform leverages the power of technology to create an engaging, personalized, and adaptive learning environment. By offering interactive problem-solving exercises, real-time progress tracking, and immersive multimedia lectures, EDU SOLVE empowers learners to strengthen their academic skills and rebuild their confidence in learning.

More than just a tool for academic improvement, EDU SOLVE embodies a broader mission to elevate the overall quality of education in the Philippines. Through the integration of modern pedagogical techniques and data-driven insights, it seeks to align with national educational reforms while addressing the unique needs of Filipino students. EDU SOLVE aims to create a learner-centered environment where students feel motivated to learn, understand foundational concepts before advancing, and build long-term academic success. Ultimately, EDU SOLVE envisions a future where Filipino learners thrive not only within their local academic environments but also on the global stage—becoming a generation equipped to compete, innovate, and succeed in an increasingly interconnected world.

1.1 Project Background

The education sector in the Philippines faces significant challenges, particularly in enhancing students' proficiency in fundamental academic subjects such as Mathematics, Science, and Reading. Despite ongoing efforts to improve the quality of education, the results of the 2022 Programme for International Student Assessment (PISA) revealed that the Philippines ranked among the lowest-performing countries globally. This highlighted not only the need for improved teaching methods but also a more engaging and effective approach to learning.

In response to these challenges, EDU SOLVE was conceptualized as an innovative educational platform aimed at addressing the academic deficiencies faced by Grade 10 students in the country. The platform is designed to bridge the learning gaps by offering interactive problem-solving exercises, engaging multimedia lectures, and real-time progress tracking tools. EDU SOLVE strives to make learning more accessible, engaging, and tailored to the needs of students, providing them with the tools necessary to improve their academic skills and boost their confidence in their educational journey.

The platform's features, including Unlockable Lectures, Progress Tracking, and Interactive Exercises, are structured to build knowledge progressively. As students advance through the lessons, they unlock additional content based on their progress, ensuring they have mastered foundational concepts before moving on to more advanced topics. This approach not only ensures a personalized learning experience but also promotes self-paced learning, enabling students to take ownership of their academic growth. The features integrate modern educational strategies such as adaptive learning and gamification, creating an engaging environment that encourages students to actively participate in their academic journey.

EDU SOLVE aims to be more than just a learning tool; it is part of a larger vision to enhance the quality of education in the Philippines, aligning with national educational reforms and helping students become globally competitive. By integrating modern educational techniques and leveraging the power of technology, EDU SOLVE supports the transformation of the Filipino education system, empowering students to succeed academically and excel in a rapidly evolving world.

1.2 Purpose and Objectives

Purpose

The purpose of EDU SOLVE is to support the academic growth of Grade 10 students in the Philippines, addressing the challenges highlighted in international assessments like PISA. This platform aims to provide an innovative, interactive, and adaptive learning environment that empowers students to improve their academic skills. Through engaging educational materials and comprehensive progress-tracking features, EDU SOLVE strives to equip learners with the tools they need to succeed in their studies and develop critical competencies. It focuses on fostering long-term academic achievement and preparing students for both national and international assessments by offering personalized learning pathways.

Objectives

- To improve the proficiency of Grade 10 students in core academic subjects such as Mathematics, Science, and Reading by offering interactive, personalized learning experiences.
- To provide students with a progressive learning path where they can track their academic progress and unlock additional learning content as they advance.
- To create an engaging and supportive environment that boosts students' confidence in learning and enables them to take ownership of their academic journey.
- To foster self-paced learning and encourage students to complete foundational tasks before advancing to more complex material, ensuring they build strong academic foundations.
- To help elevate the Philippines' education system by providing a modern, data-driven tool that aligns with national educational reforms and supports student success on a global scale.

1.3 Significance

The significance of EDU SOLVE extends far beyond being a mere educational platform; it is a catalyst for transforming the landscape of education in the Philippines. By addressing the fundamental academic challenges faced by students, particularly in core subjects like Mathematics, Science, and Reading, EDU SOLVE provides students with the skills and knowledge necessary to thrive academically. Through its interactive tools, personalized content, and measurable progress tracking, EDU SOLVE supports students in taking control of their learning journey.

This initiative is particularly critical in light of global education assessments such as PISA, where Filipino students have shown room for improvement. EDU SOLVE helps bridge this gap by offering an engaging and user-friendly learning experience that motivates students to succeed. By focusing on the core subjects that are crucial to a student's academic development, the platform allows learners to build confidence, strengthen their knowledge, and ultimately perform better in both local and international assessments.

Furthermore, EDU SOLVE contributes to the broader goal of improving the Philippines' standing in global education rankings. By providing students with modern learning tools and strategies, EDU SOLVE helps raise the bar for education in the country, ensuring that Filipino students are better equipped to face future academic and professional challenges. In doing so, it not only supports individual success but also contributes to the overall growth of the nation's educational system, preparing Filipino students to compete on the global stage.

1.4 Target Audience

The primary target audience for EDU SOLVE includes Grade 10 students in the Philippines who are struggling in key academic areas such as Mathematics, Science, and Reading. The platform is specifically designed to help these students improve their proficiency in these subjects, providing a personalized, engaging, and adaptive learning experience that helps them progress at their own pace.

Stakeholders include:

- **CHED (Commission on Higher Education):** Implementing EDU SOLVE as part of their efforts to improve educational standards across the country.
- **WMSU (Western Mindanao State University) Faculties:** Using EDU SOLVE as a supplementary tool in the classroom to enhance learning and help students improve their academic skills.

- **Educators and Parents:** Teachers and parents who are invested in supporting students' academic development and ensuring that they have access to the resources needed to succeed.

Additionally, EDU SOLVE can serve students from different learning backgrounds, including those who benefit from a more structured, progressive learning approach. It is designed to cater to the needs of a wide range of learners, helping them take ownership of their education and giving them the tools to succeed academically.

CHAPTER 3

SYSTEM FEATURES

3.1. Interactive Problem-Solving Exercises

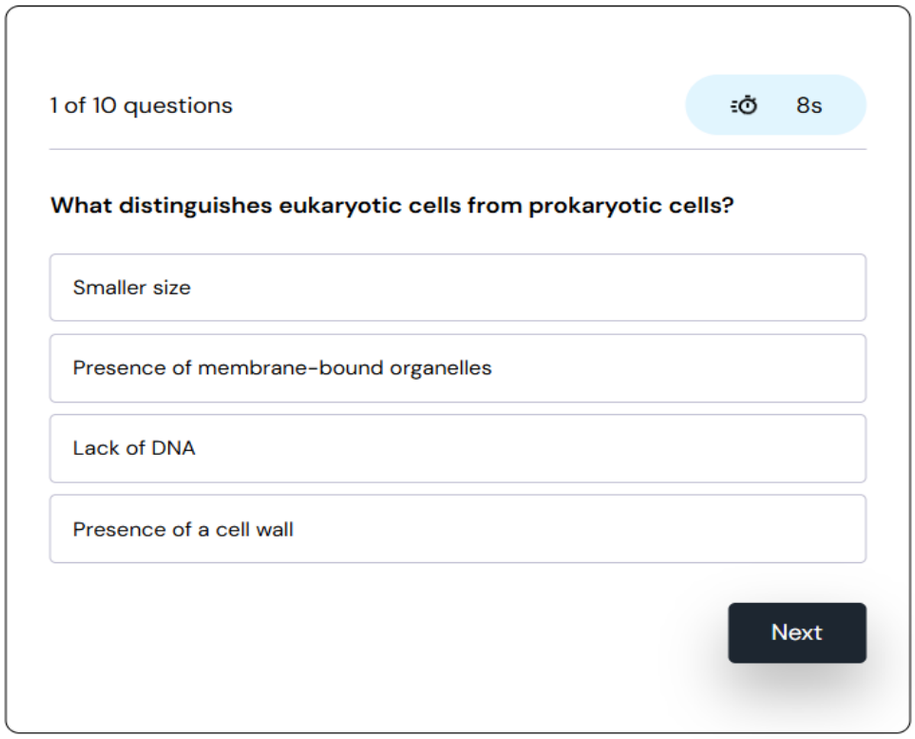
Our interactive problem-solving exercises are designed to enhance learning through active engagement, offering a hands-on approach to mastering core subjects. These exercises are tailored to improve critical thinking and problem-solving skills by presenting real-world scenarios and guiding users through step-by-step solution processes.

In addition to the exercises, we offer **Practice Tests** and **Final Tests** to assess progress and mastery of the material. The practice tests allow students to test their knowledge in a low-pressure environment, while the final tests evaluate overall understanding and readiness to move forward.

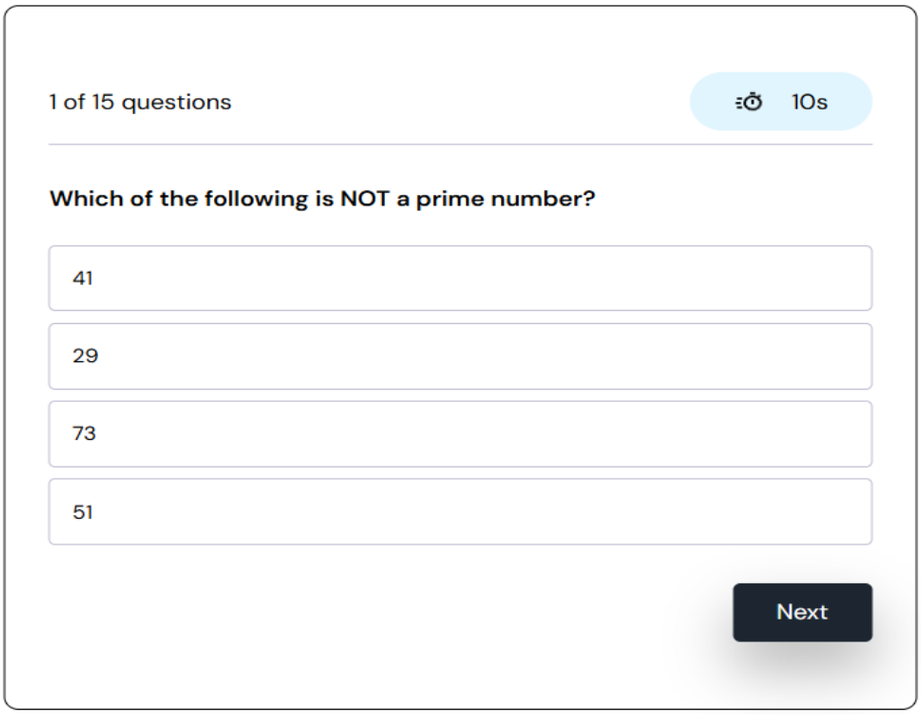
Key Features:

- **Practice Tests:**
  These tests provide users with an opportunity to apply what they have learned in a simulated exam format. By reinforcing knowledge through hands-on exercises, the practice tests help boost students' confidence and deepen their understanding of foundational concepts.

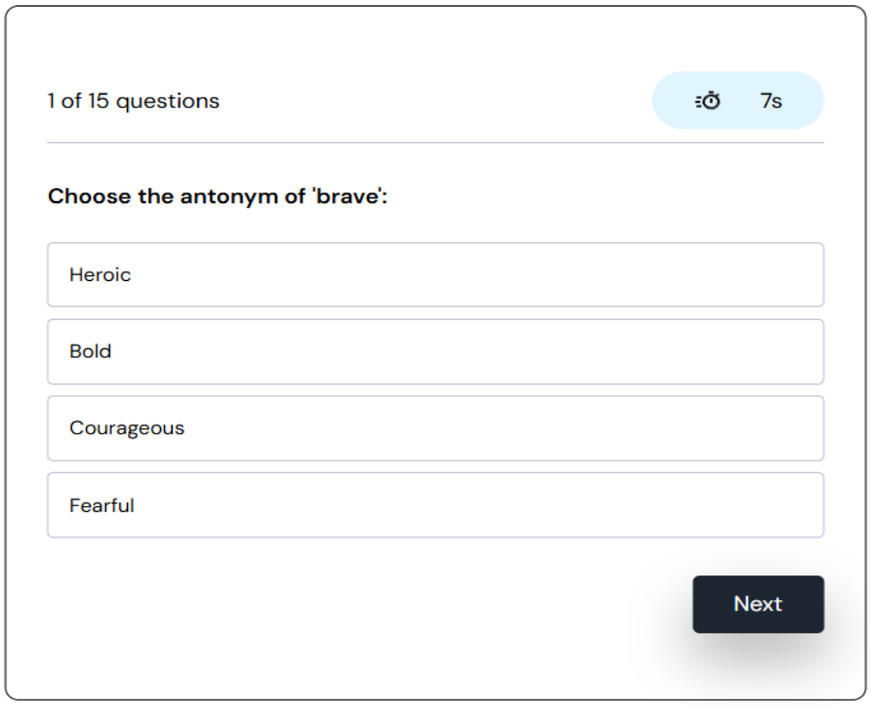
- ***Visual Representation:***
  Please see Figures 3.1.1a, 3.1.1b, and 3.1.1c for subject-specific views of the Practice Tests.
  These images illustrate the distinct layouts for Mathematics, Science, and Reading modules, showcasing how the system adapts problem-solving exercises to the unique requirements of each subject.



*Please see **Figure 3.1.1:** Overview of Interactive Problem-Solving Exercises (Practice Test) across Science, Mathematics, and Reading modules.*



*Please see **Figure 3.1.1:** Overview of Interactive Problem-Solving Exercises (Practice Test) across Science, Mathematics, and Reading modules.*



*Please see **Figure 3.1.1:** Overview of Interactive Problem-Solving Exercises (Practice Test) across Science, Mathematics, and Reading modules.*

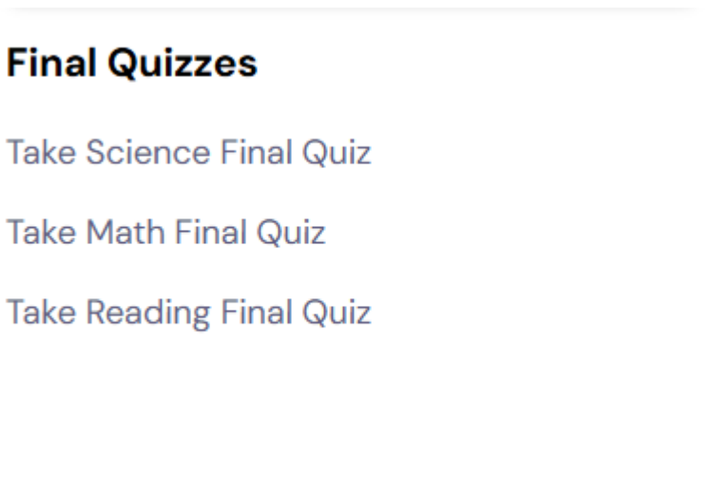
- **Final Tests:**

  The final tests are designed as a comprehensive evaluation tool, assessing students' mastery of the subject matter. These tests determine their readiness to advance to the next level and tackle more complex applications in Mathematics, Science, and Reading.

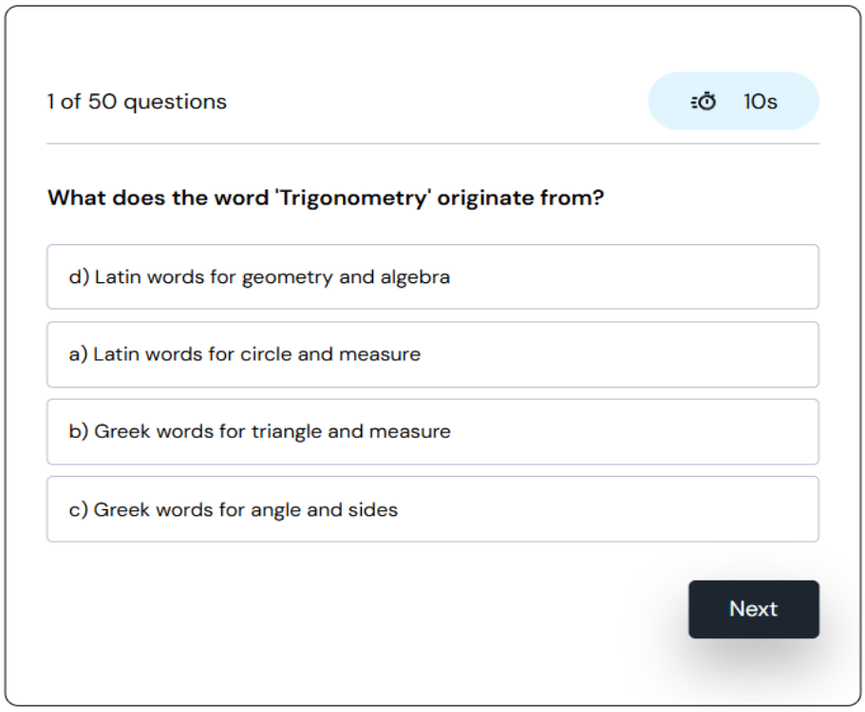
  - *Visual Representation:*

    Please see Figures 3.1.2a, 3.1.2b, and 3.1.2c for subject-specific views of the Final Tests.

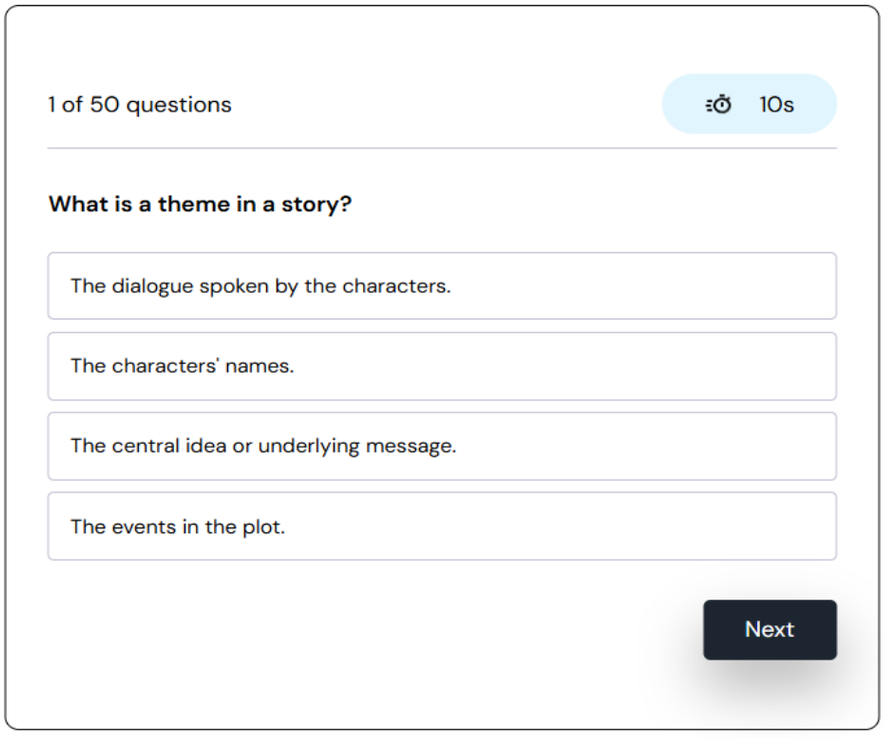
    These images demonstrate the structured and challenging layouts for Mathematics, Science, and Reading modules. They highlight the advanced problem-solving scenarios designed to assess comprehensive understanding and prepare learners for more complex academic tasks.



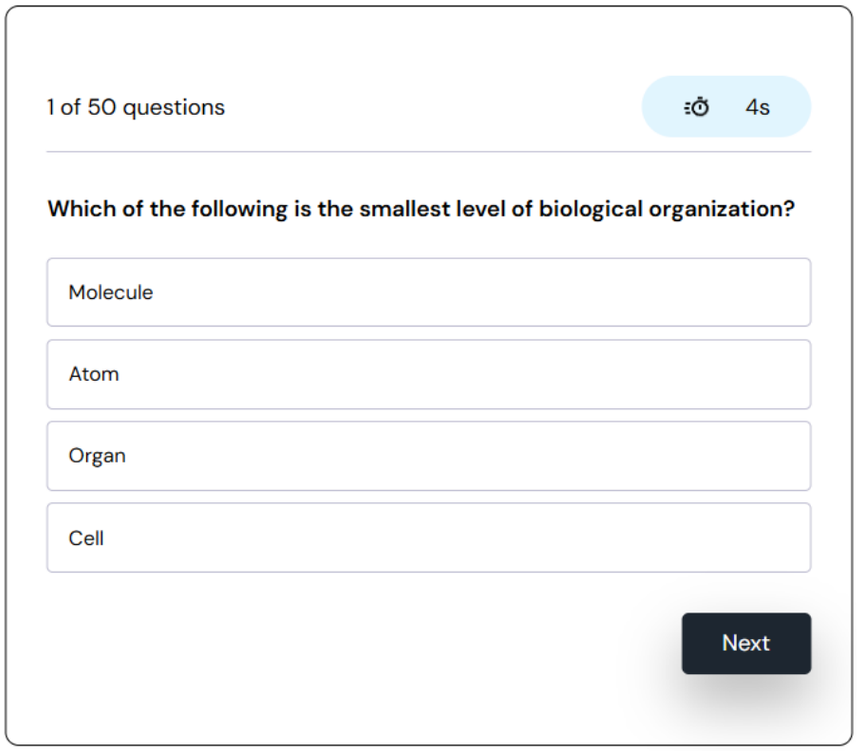
*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*



*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*



*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*



*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*

3.2 Engaging Lectures

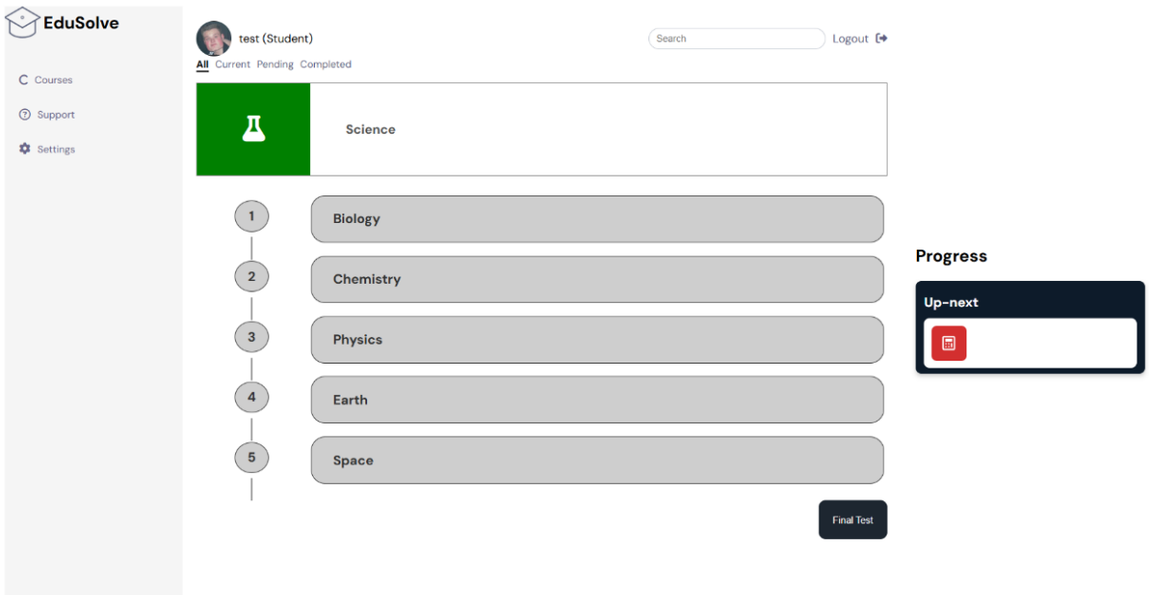
Our engaging lectures offer a comprehensive and interactive learning experience across core subjects: **Science, Mathematics, and Reading**. Through a blend of multimedia resources and dynamic presentations, these lectures are designed to spark curiosity, foster critical thinking, and make learning an enjoyable process.

Key Features:

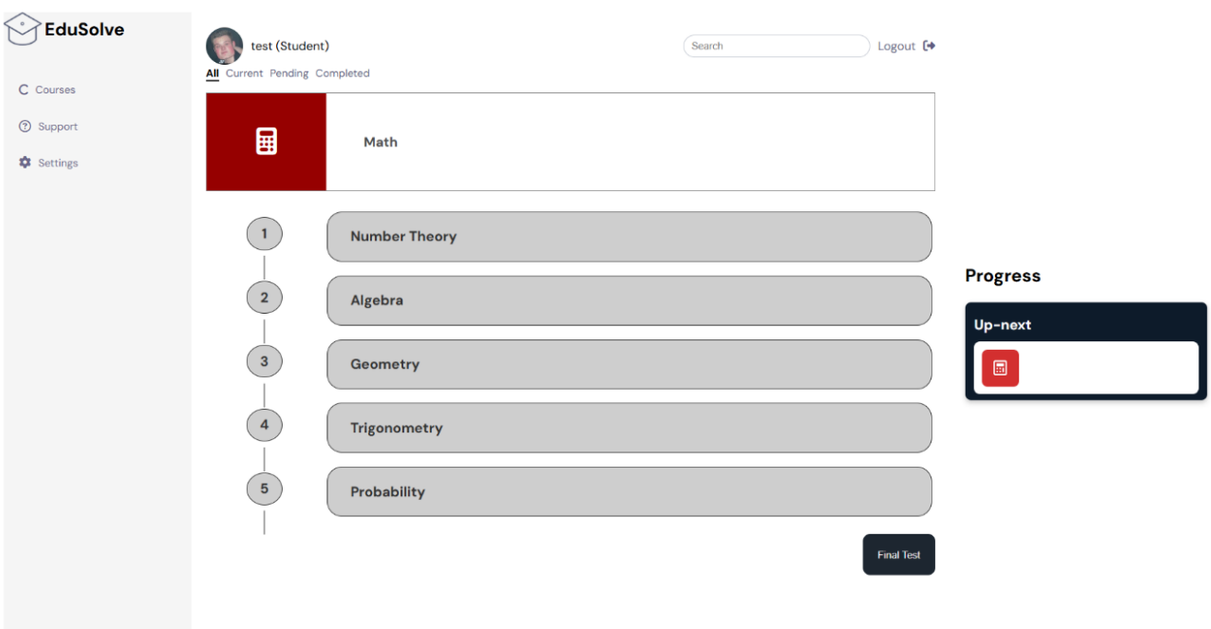
- **Science**: Explore the wonders of the natural world through captivating experiments, virtual field trips, and data-driven analysis.
- **Mathematics**: Master essential mathematical concepts with interactive problem-solving, real-world applications, and engaging visual aids.
- **Reading**: Develop strong language skills through engaging storytelling, creative writing exercises, and interactive discussions.

By combining traditional lecture formats with innovative digital tools, our lectures offer a truly immersive and effective learning experience.

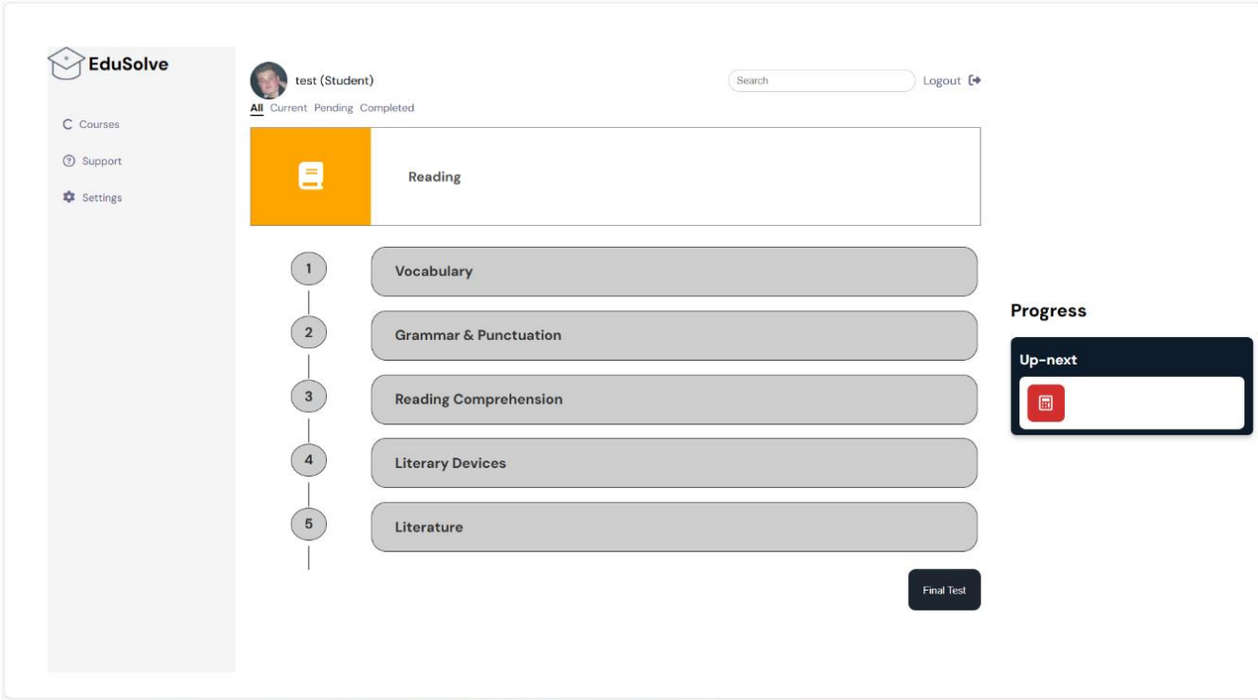
Please see **Figure 3.2.1** for an overview of the engaging lectures, illustrating the structured layout and interactive elements within each subject area.



Please see Figure 3.2.1 for an overview of the Science modules. This image illustrates the structured layout and ease of navigation within the Science subject area.



Please see Figure 3.2.2 for an overview of the Mathematics modules. This image illustrates the structured layout and ease of navigation within the Mathematics subject area.



Please see Figure 3.2.3 for an overview of the Reading modules. This image illustrates the structured layout and ease of navigation within the Reading subject area.

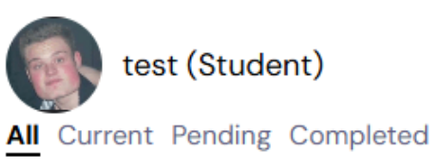
3.3 Progress Tracking

Performance Tracking:

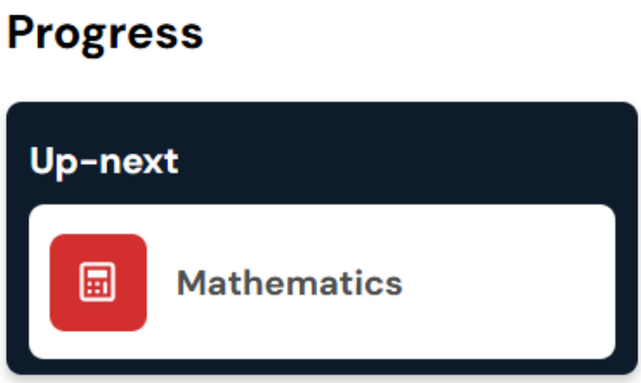
The Edu Solve system continuously tracks student performance, providing a dynamic view of their progress across various subjects. This feature ensures that students are assessed at multiple points in their learning journey, enabling real-time insights into their strengths and areas requiring improvement.

- **Progress Bars:**
  One of the key features of the performance tracking system is the use of **progress bars**. These visual indicators display the current status of a student's progress in a subject or module. The bars are segmented into three distinct states:
  - **Current:** This section of the progress bar shows how far the student has progressed in the current topic or module. It provides a visual representation of the work already completed.
  - **Pending:** This section indicates the remaining content or tasks yet to be completed. It shows students which areas are still pending, giving them a clear roadmap of what's left to cover.
  - **Completed:** Once a student finishes a module or task, the "completed" section fills in, signifying that they have successfully finished that portion of their learning. This provides a sense of accomplishment and motivates students to continue their studies.

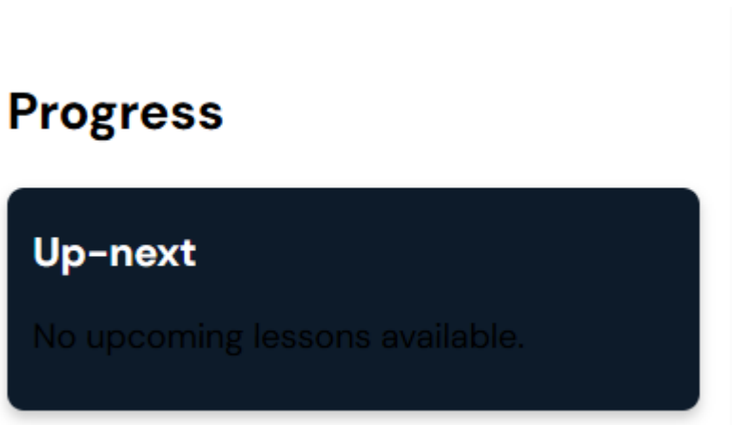
Please see Figure 3.3.1 for a sample view of the progress bar, which illustrates how the current, pending, and completed sections visually represent student progress within the Edu Solve system.



*Please see Figure 3.2.*1 for a sample view of the progress bar



Please see Figure 3.2.1 for a sample view of the progress bar



Please see Figure 3.2.1 for a sample view of the progress bar

3.4. Unlockable Lectures

The **Unlockable Lectures** feature within EDU SOLVE is designed to provide students with an engaging and progressive learning experience. It allows students to access educational content in a structured manner, where certain lessons or lectures become available based on their progress in earlier sections. The system is designed to help students unlock new material as they complete previous tasks, ensuring they build knowledge step-by-step.

Please refer to **Figure 3.4.1** for a sample view of the **Unlockable Lectures**, which illustrates how the current, pending, and completed sections visually represent student progress within the Edu Solve system.

Key Features of Unlockable Lectures:

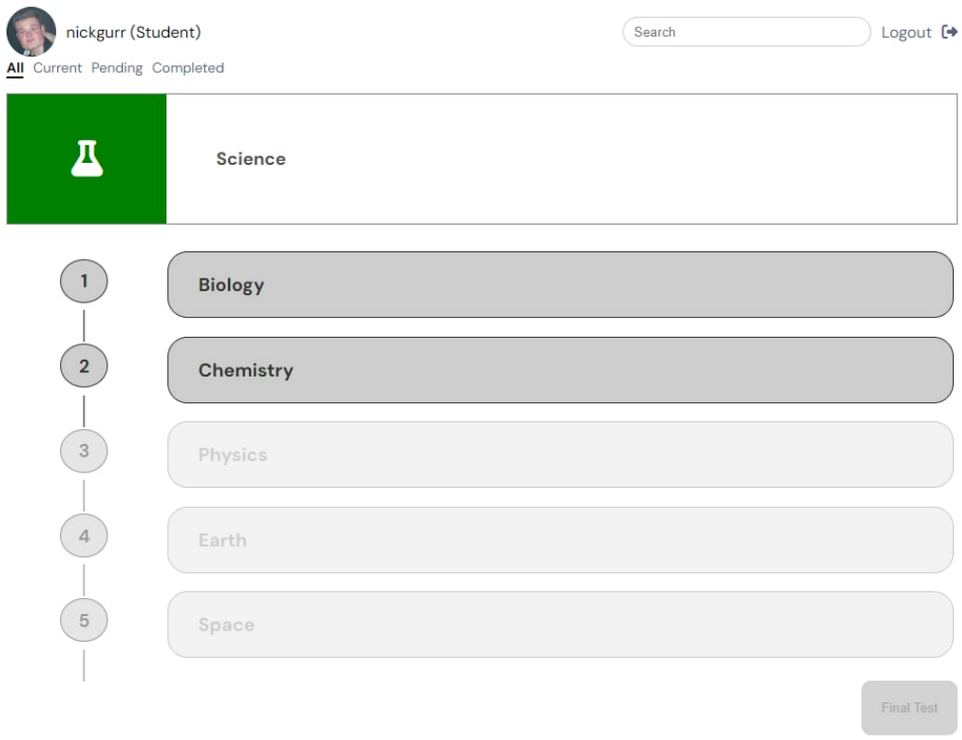
- **Current Lectures:**
  These are the lectures that students are actively working on. They are typically unlocked based on the completion of prior sections or based on predefined milestones within the curriculum. The current section is marked as "in progress" and may include interactive exercises, video content, or reading material.

- **Pending Lectures:**
  Pending lectures are those that will become available once the student successfully completes the current section. These lectures are locked and cannot be accessed until the student progresses through the prerequisite content. Pending lectures are visually indicated in the system with a "locked" status, ensuring students understand that further progress is required.

- **Completed Lectures:**
  Once a student has finished a lecture, it is marked as "completed" in the system. The completed section is highlighted to reflect the student's academic progress. This helps students track their learning journey and see how much content they have already mastered. Completed lectures also serve as a foundation for unlocking the next set of lectures.

  The unlocking mechanism provides a clear structure for students, motivating them to complete tasks before moving on to more advanced content. This approach not only ensures students grasp foundational concepts before advancing but also encourages continuous learning and achievement.

  By structuring content in this manner, EDU SOLVE helps students progress at their own pace, keeping them engaged, challenged, and motivated throughout their learning journey.



*Please see Figure 3.4.*1 for a sample of the unlockable lectures

3.6 Account Management

The Account Management feature enhances the system's usability and security by allowing users to update their account credentials directly within the platform. This feature ensures that users can manage their personal information securely and efficiently.

Key Features:

- **Update Username:** Users can change their username to reflect their preferences or correct errors during registration.
- **Update Password:** Users can enhance security by updating their passwords periodically.

Workflow:

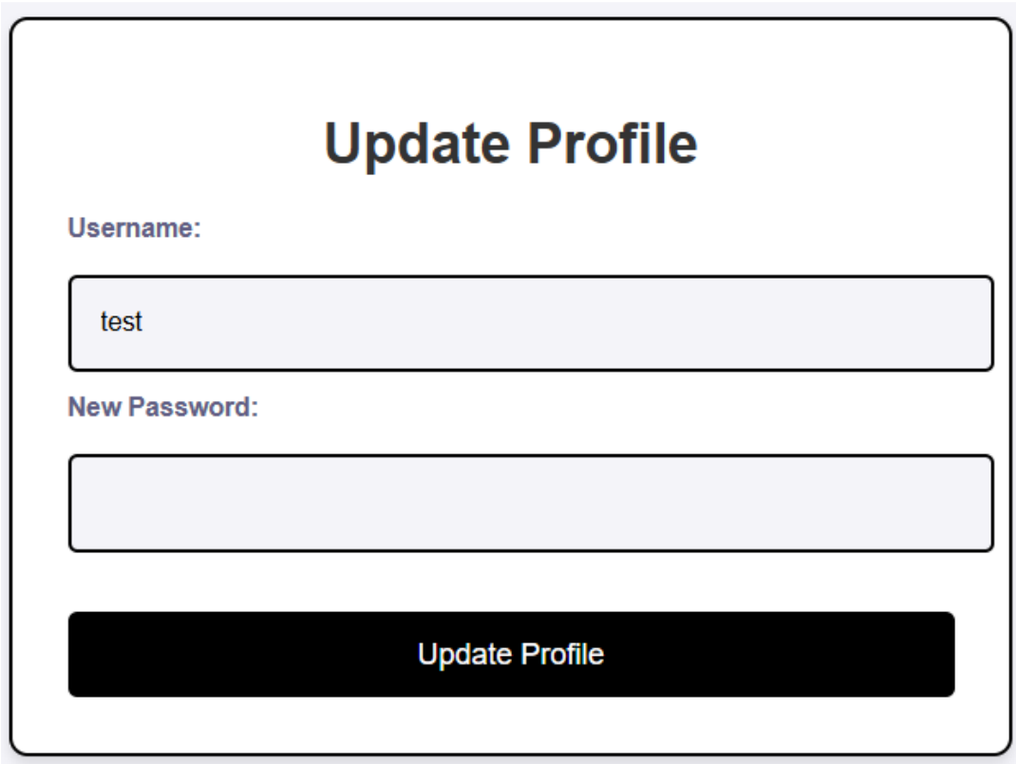
1. Navigate to the **Settings** menu.
2. Select either **Update Username** or **Update Password**.
3. Input the required information in the respective fields.
    - For username updates: Enter the new username.
    - For password updates: Enter the current password and the new password (with confirmation).
4. Submit the changes. The system validates the inputs and applies the updates upon successful verification.

Benefits:

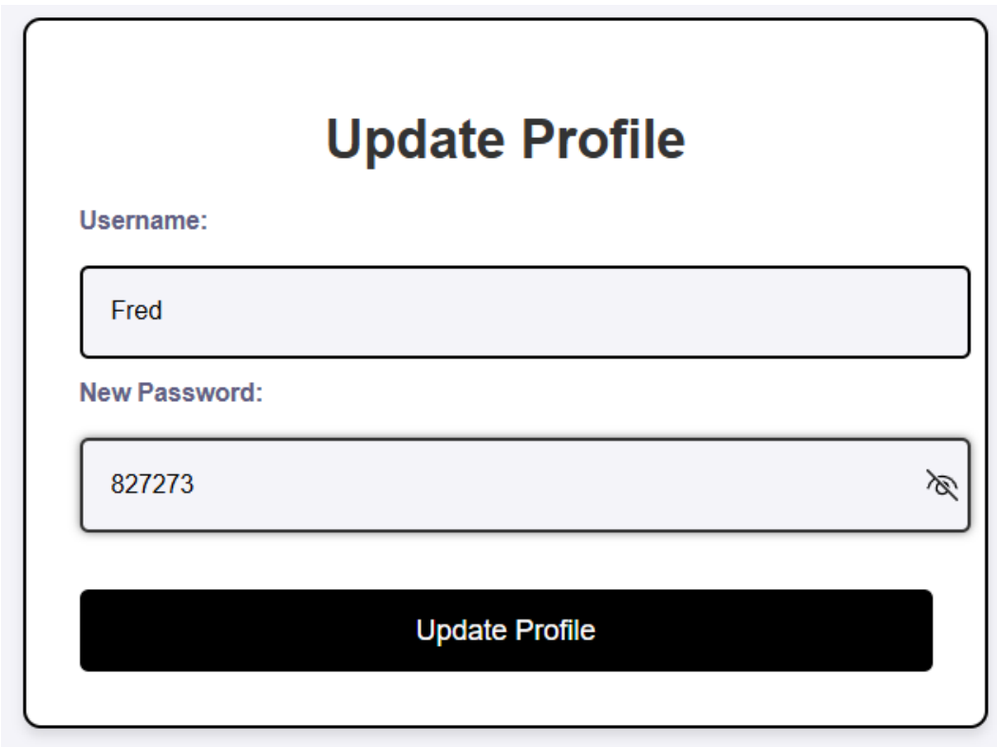
- **Improved Security:** Encourages users to keep their credentials secure and updated.
- **Ease of Use:** A simple and intuitive process ensures all users, regardless of technical ability, can manage their accounts effectively.
- **Enhanced User Control:** Gives users the ability to personalize and secure their accounts.

Figure Reference:

Please see **Figure 3.6.1** and **Figure 3.6.2** for an example of the Settings interface, showcasing the options for updating username and password.



*Please see Figure 3.6.*1 for a sample of the Settings interface



*Please see Figure 3.6.*2 for a sample of the update Settings interface

CHAPTER 4

SYSTEM ARCHITECTURES

4.1 System Workflow

The system workflow for **Edu Solve** begins with user interaction on the front end. Users access the website to navigate through educational content, perform actions like logging in, submitting forms, or viewing resources. These interactions trigger processes in the back end, which handles data processing, authentication, and communication with the database. The workflow ensures a seamless exchange of information between the user interface, application logic, and database storage.

4.2 Front-End Architecture

The front end of Edu Solve is designed with a focus on providing a seamless and engaging user experience. The architecture prioritizes responsiveness, interactivity, and ease of navigation. It uses a combination of core technologies and features to ensure that users have a smooth and effective learning experience.

Technologies Used

1. **HTML (HyperText Markup Language)**
    - Serves as the backbone of the website, structuring the content and elements.
    - Key Implementation:
        - Used for creating static elements like login forms, subject lists, and quiz pages.
        - Example: Structuring input fields for user registration, quiz questions, and content display areas.
2. **CSS (Cascading Style Sheets)**
    - Enhances the visual appeal of the platform while ensuring responsiveness across devices.
    - Key Implementation:
        - Designed a consistent color scheme and typography aligned with the educational theme.
        - Applied media queries for responsive design, adapting layouts for mobile, tablet, and desktop screens.

        - Example: The dashboard grid layout adjusts based on screen size—displaying 3 columns on desktops and stacking them vertically on smaller devices.

3. **JavaScript**
    - Adds dynamic behavior and interactivity to the website, making the learning experience more engaging.
    - Key Implementation:
        - Form Validation: Ensures user input is correct, such as email format validation on registration forms.
        - Interactive Features: Powers dynamic quizzes, dropdown menus, and real-time progress tracking.
        - Example: Real-time quiz result updates and filtering subjects based on user selection.

Core Features of the Front End

1. **Responsive User Interface**

- Optimized for laptops and desktop computers.
- Example: The navigation menu is designed for larger screens, ensuring a clear and accessible layout for users on desktop and laptop devices.
- Please see Figure 4.1 for the design details.

2. **Interactive Components**
    - Implements engaging features like:
        - Progress Bars: Show users' progress during quizzes.
        - Pop-ups and Modals: For notifications, password resets, or providing hints during quizzes.
        - Example: A modal appears after completing a quiz, showing performance results.

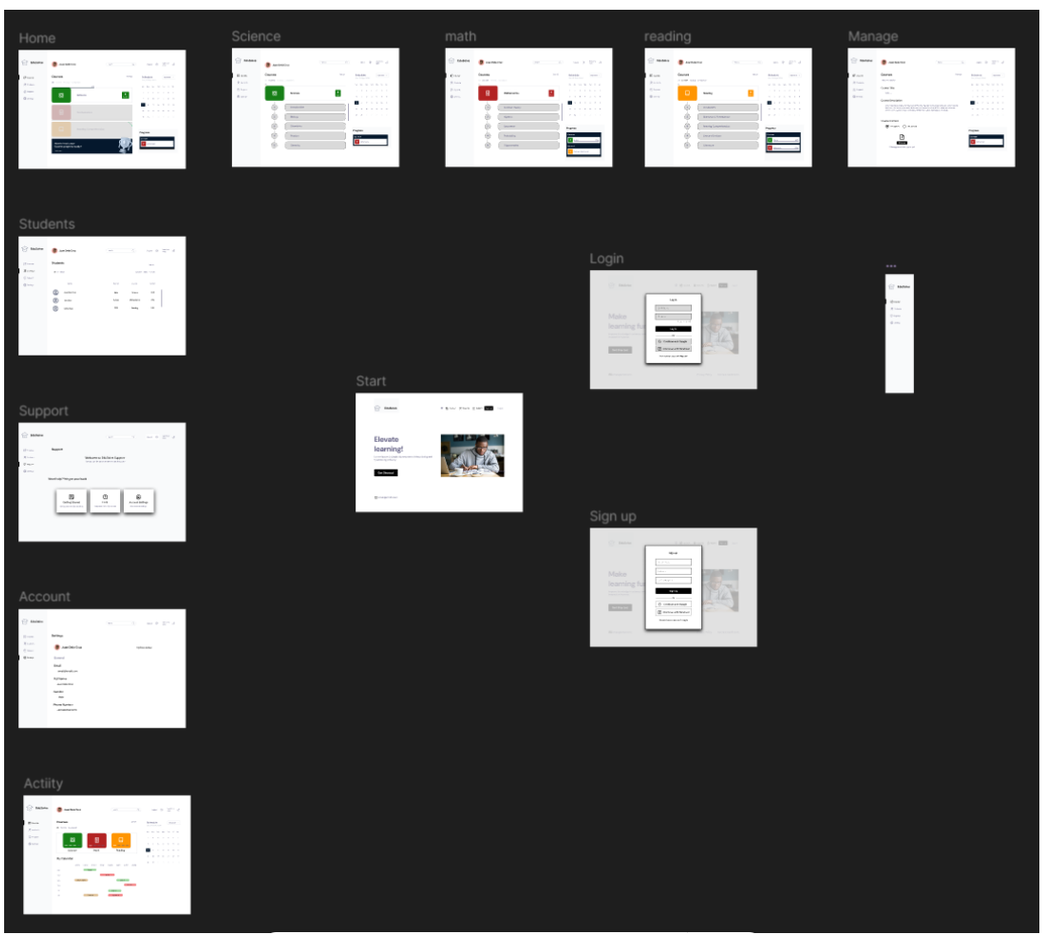
3. **Seamless Communication with the Back End**

   - The front end communicates with the back end using AJAX, ensuring smooth real-time updates without page reloads.
   - Example: When users unlock a new subject, the front end sends a request to the back end, and the updated list of subjects is displayed dynamically

4. **Prototyping and Design**

   - Figma was used for prototyping and refining the user interface before actual implementation.
     - Example: The dashboard layout, quiz screens, and onboarding flow were tested and iterated in Figma to ensure usability and consistency**.**

Please see Figure 4.1 for the initial wireframe design the following figures for the wireframes and prototypes that demonstrate the user interface and flow:



Please see Figure 4.1 for the initial wireframe design. This image illustrates the layout and flow of the front-end interface, showcasing how users interact with the system from login to dashboard.

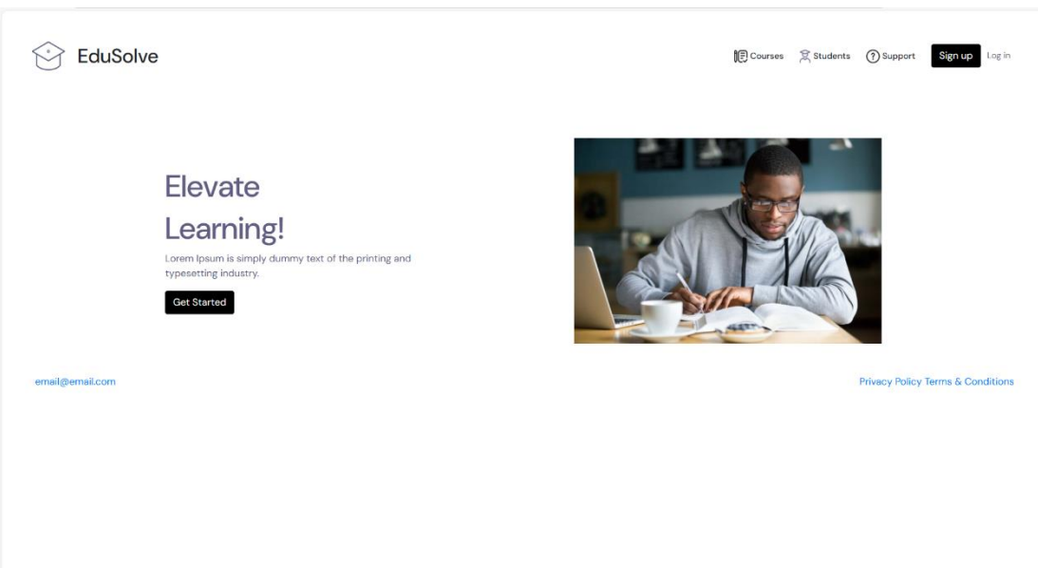
User-Centric Design Principles

The front-end architecture follows a user-centric approach, ensuring the following:

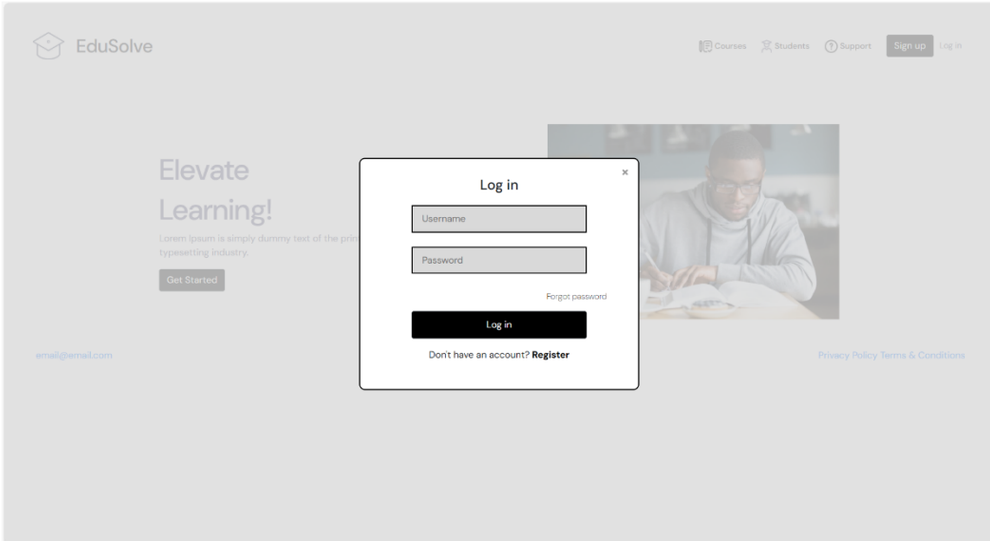
- **Intuitive Navigation:** Users can quickly access their assigned subjects and switch between dashboards, quizzes, and progress trackers.
- **Seamless Flow:** Clear, responsive transitions between pages (e.g., Login → Dashboard → Progress Tracking) guide users through the app.
- **Visual Feedback:** Provides clear feedback to users, such as quiz completion bars or form error highlights.
    - Example: When a quiz is completed, users see a completion bar and score feedback.

Visuals:

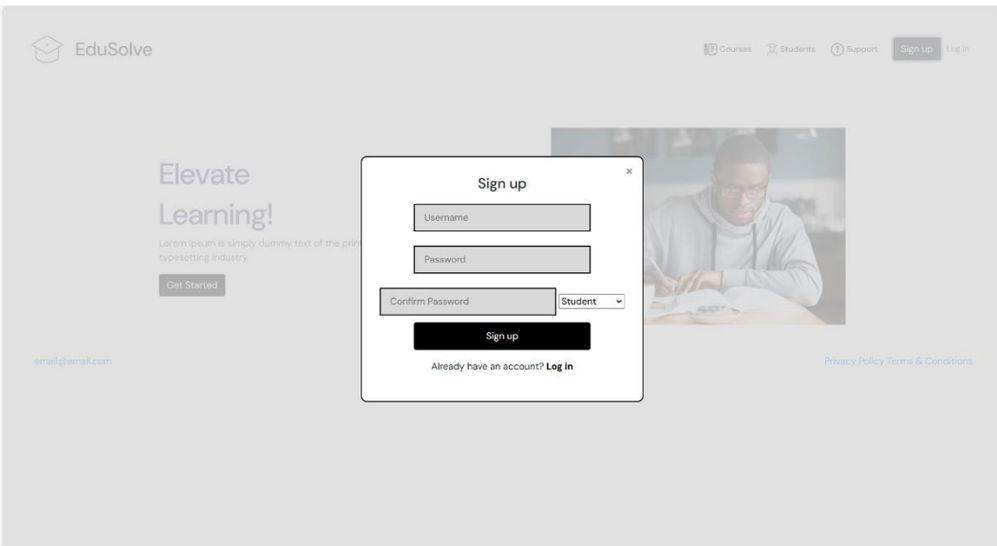
Please see the following figures for the wireframes and prototypes that demonstrate the user interface and flow:



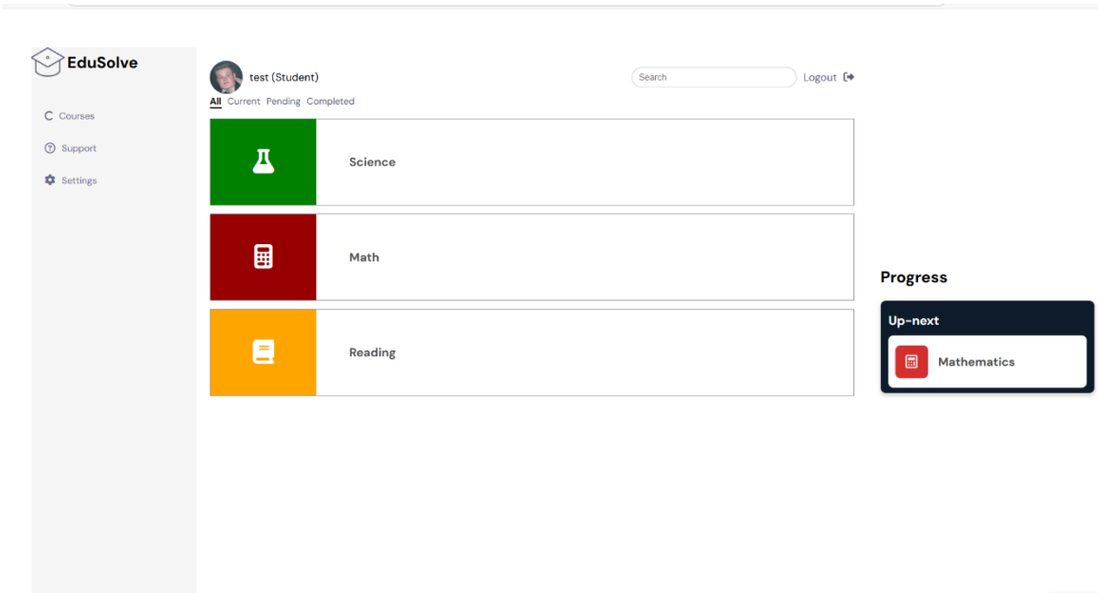
- Figure 4.1: *Homepage - The entry point of the system, where users can initiate their journey and log in or sign up.*



- Figure 4.2: *Login Page - Where users input their credentials to access the system*.



- Figure 4.3: *Sign Up Page - New users can register for the system here*.



- Figure 4.4: *Dashboard - The main interface users see after logging in, where they can access their subjects, progress, and other features*.

These images showcase how the interface elements are arranged and how users interact with them.